

Domain 9: Infrastructure Vulnerabilities

Operational Context I

Smart grid agents balance load and generation in real-time. **FR1 = Maintain grid frequency at 60Hz. FR2 = Prevent equipment damage (Overload Protection).**

Adversary Classification: Ω_2 (Peripheral) — the adversary cannot modify the grid control software directly, but can inject false sensor telemetry through compromised IoT devices in the network periphery [?].

Axiomatic Design Formulation I

The system has two functional requirements, yielding a 2×2 Design Matrix [?]:

Uncoupled (pre-attack) Design Matrix. Under normal operation, the design is uncoupled:

$$\begin{Bmatrix} FR_1 \\ FR_2 \end{Bmatrix} = \begin{bmatrix} A_{11} & 0 \\ 0 & A_{22} \end{bmatrix} \begin{Bmatrix} DP_1 \\ DP_2 \end{Bmatrix} \quad (1)$$

where DP_1 = Frequency Regulation Controller and DP_2 = Overload Protection (Load Shedding) Controller. Each FR is independently satisfied by its corresponding DP.

Post-attack (coupled) Design Matrix. Sensor masquerading introduces off-diagonal coupling:

$$\begin{Bmatrix} FR_1 \\ FR_2 \end{Bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & -A_{22} \end{bmatrix} \begin{Bmatrix} DP_1 \\ DP_2 \end{Bmatrix} \quad (2)$$

Axiomatic Design Formulation II

The sign reversal $A_{22} \rightarrow -A_{22}$ represents **FR Polarity Inversion**: the “Prevent Damage” function is weaponized to *cause* damage (blackout via unnecessary load shedding). The off-diagonal terms A_{12}, A_{21} represent the adversary-induced coupling between frequency regulation and overload protection, violating the Independence Axiom [?, ?].

The Goal Hijacking Attack I

Attackers use “Sensor Masquerading” to hijack the “Load Shedding” FR [?, ?].

- ▶ **Mechanism:** A compromised IoT botnet injects “High Load” telemetry into the grid controller, while simultaneously suppressing “Generation” readings.
- ▶ **Hijack:** The agent is forced to execute its “Emergency Load Shedding” FR to “save” the grid from a phantom overload. The goal “Prevent Damage” is weaponized to cause a blackout.

The Goal Hijacking Attack II

The threat is grounded in documented state-level operations. The Volt Typhoon campaign [?]¹—a PRC state-sponsored persistent access operation targeting U.S. critical infrastructure—maintained undetected presence in operational technology (OT) networks for nearly a year (February–November 2024 at Littleton Electric Light and Water Departments). This demonstrated precisely the kind of prolonged Ω_2 positioning that would enable the sensor masquerading attack described above. The scale of the vulnerability is significant: 2,451 ICS vulnerabilities were disclosed between December 2024 and November 2025 from 152 vendors, internet-exposed ICS devices increased by 40%, and ransomware attacks on industrial targets rose 355% between 2020 and 2025.

OODA Loop Transients I

The attack propagates through the OODA loop [?] as follows:

1. **Observe:** Agent sees “Load > Capacity” (False Data).
2. **Orient:** Emergency State triggered. This is the primary target phase — false sensor data corrupts the agent’s situational awareness, causing it to misclassify a stable grid as critically overloaded.
3. **Decide:** Cut power to Sector 7 (Hospital District).
4. **Act:** Blackout occurs; the grid was stable, but the *agent* was destabilized.

CIF Defense: Behavioral Invariants, Belief Sandboxing, and Drift Detection I

CIF implements **Behavioral Invariants** (Paper 1, Def. 5.5) [?] with a partially novel extension: *physics-informed invariants* that encode conservation laws (Kirchhoff's Laws) as runtime predicates [?]. Rather than learning invariants from data (which can be poisoned), these invariants are derived from first-principles physics and cannot be overridden by any data-driven model.

CIF also implements **Belief Sandboxing** (Paper 1, Def. 5.3) [?] by isolating the emergency response pathway: before executing load shedding, the agent evaluates the emergency hypothesis in a sandboxed belief state that cross-references multiple independent sensor channels.

Finally, CIF implements **Drift Detection** (Paper 1, Def. 5.6) [?] via temporal damping that filters fast synthetic transients characteristic of cyber-attacks.

CIF Defense: Behavioral Invariants, Belief Sandboxing, and Drift Detection II

- ▶ **Conservation of Energy (Physics-Informed Behavioral Invariant):** The agent checks if the reported “High Load” is physically consistent with the current flow at the substations (Kirchhoff’s Laws). If $\sum I_{in} \neq \sum I_{out}$ beyond noise margins, the sensor data is rejected. These physics-informed invariants provide an unforgeable ground truth that no adversarial data injection can circumvent [?, ?].
- ▶ **Slow-Transient Filter (Drift Detection via Temporal Damping):** The “Emergency Shed” FR has a built-in temporal damper. It requires the overload condition to persist for $> \Delta t$ (defined by thermal limits) before acting, filtering out fast, synthetic OODA transients that are characteristic of cyber-attacks. This temporal requirement exploits the fundamental asymmetry between real thermal events (which evolve on physical timescales) and injected data (which can appear instantaneously) [?].

Summary I

Element	Value
Functional Requirements	FR_1 : Maintain grid frequency at 60Hz; FR_2 : Prevent equipment damage
Design Parameters	DP_1 : Frequency Regulation Controller; DP_2 : Overload Protection Controller
Attack Vector	Sensor masquerading via compromised IoT botnet injecting false telemetry
Adversary Class	Ω_2 (Peripheral)
OODA Target Phase	Orient
Attack Pattern	FR Polarity Inversion (“Prevent Damage” weaponized to cause blackout)

Summary II

Primary CIF Defense

Behavioral Invariants
(physics-informed) + Belief
Sandboxing + Drift
Detection (temporal
damping)

Novel Contribution

Physics-informed invariants
encoding conservation laws
as unforgeable runtime
predicates
